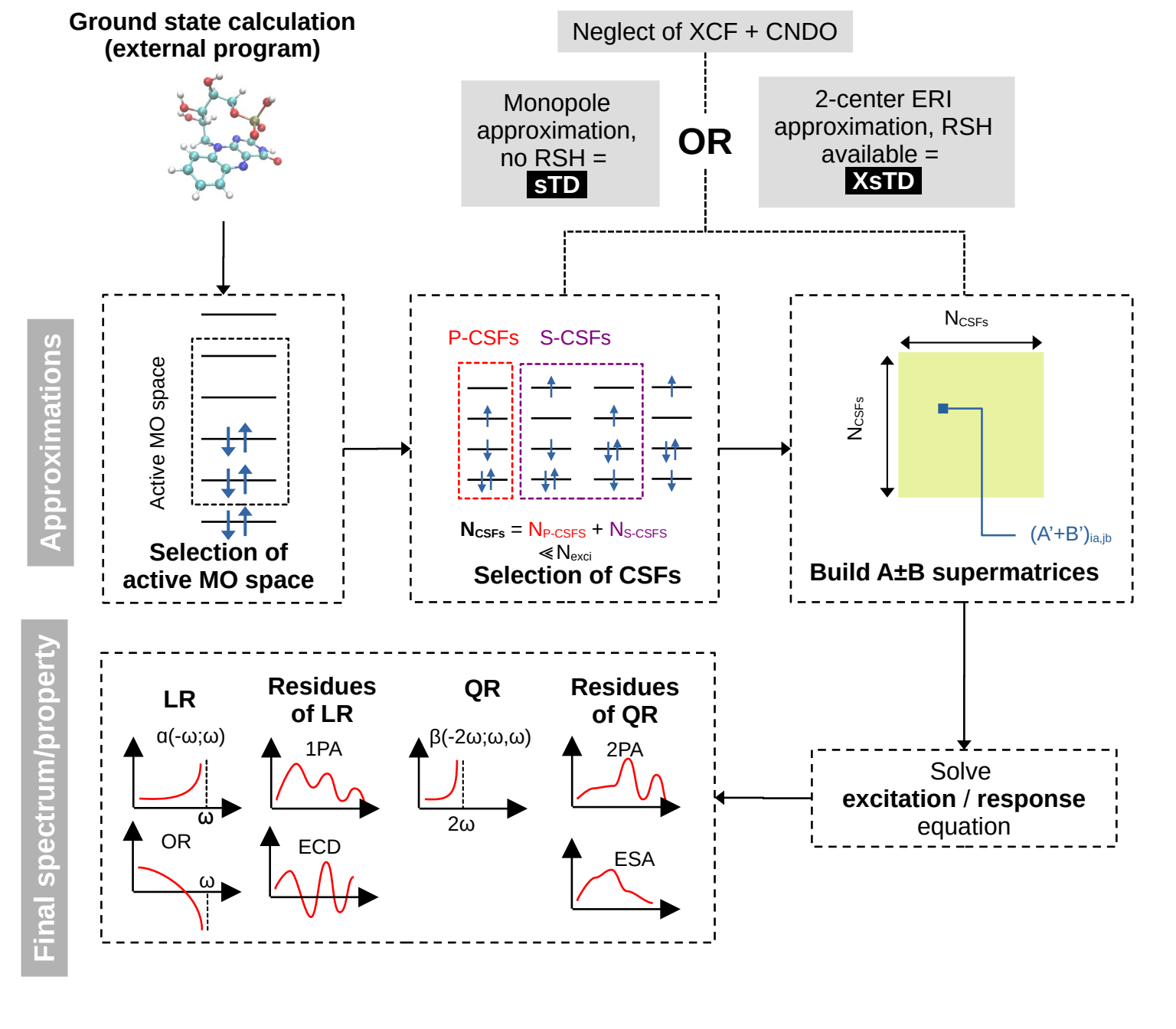
Ground state calculation
(external program)
Neglect of XCF + CNDO
Monopole approximation, no RSH = sTD
OR
2-center ERI approximation, RSH available = XsTD
Approximations
Active MO space
Selection of active MO space
P-CSFs
S-CSFs
NCSFs = NP-CSFS + NS-CSFS
≪ Nexci
Selection of CSFs
NCSFs
NCSFs
(A'+B')ia,jb
Build A±B supermatrices
Final spectrum/property
LR
α(-ω;ω)
ω
OR
ω
Residues of LR
1PA
ECD
QR
β(-2ω;ω,ω)
2ω
Residues of QR
2PA
ESA
Solve excitation / response equation